PRML Assignment 3: Air Quality Forecasting Report

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2026 Spring

Abstract

This assignment studies one-hour-ahead PM2.5 forecasting with the Beijing Air Quality dataset. The raw data contains hourly pollution, meteorological variables, and wind direction from 2010 to 2014. The forecasting problem is converted into supervised learning by using the previous 24 hours of multivariate observations to predict the next PM2.5 value. A persistence baseline, Ridge regression, Random Forest, Gradient Boosting, a multilayer perceptron, and a PyTorch LSTM are evaluated on a chronological hold-out set. The best model is LSTM, with RMSE=23.060, MAE=12.339, and R2=0.939.

1. Dataset and Task Definition

The dataset records hourly weather and pollution conditions. Its variables include PM2.5 concentration, dew point, temperature, pressure, wind direction, wind speed, cumulative snow hours, and cumulative rain hours. The target is the pollution concentration in the next hour. Because future values must not leak into training, the data is split chronologically: the first 80% of rows are used for training and the final 20% are reserved for testing.

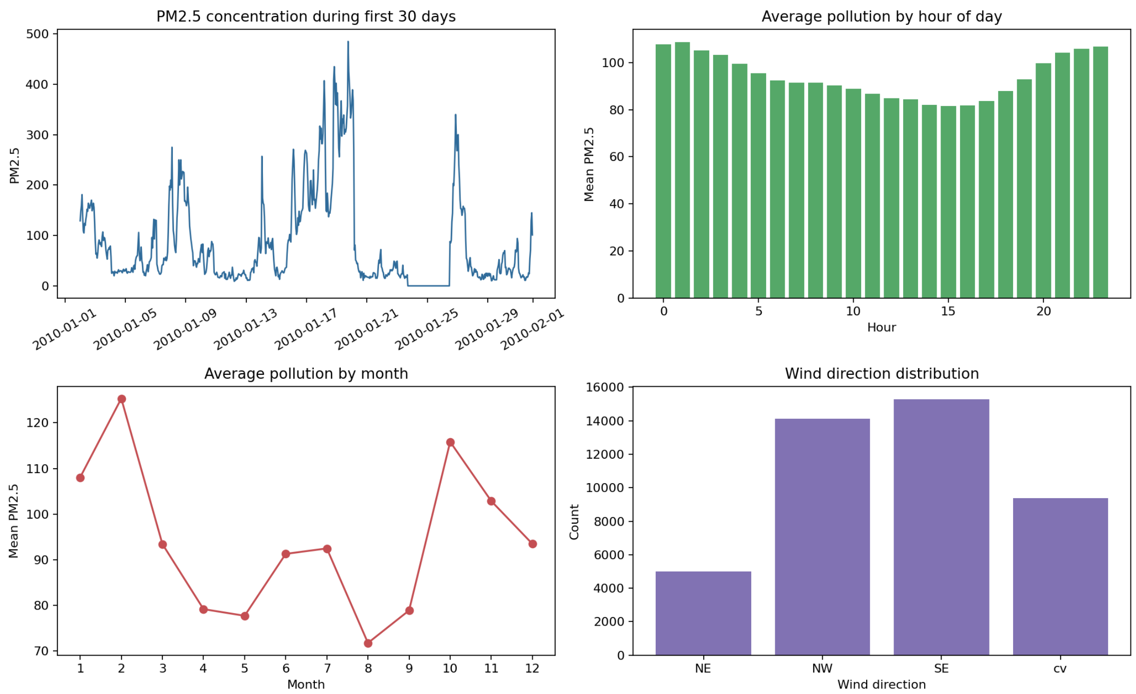

Figure 1: Exploratory views of PM2.5, hourly/monthly patterns, and wind direction counts.

2. Feature Engineering

For each target hour, the previous 24 rows are used as one lag window. Numeric variables are standardized using only the training period. Wind direction is one-hot encoded with categories learned from the training data. Traditional regressors receive the window as a flattened feature vector, while the LSTM receives the same data as an ordered 24-step sequence.

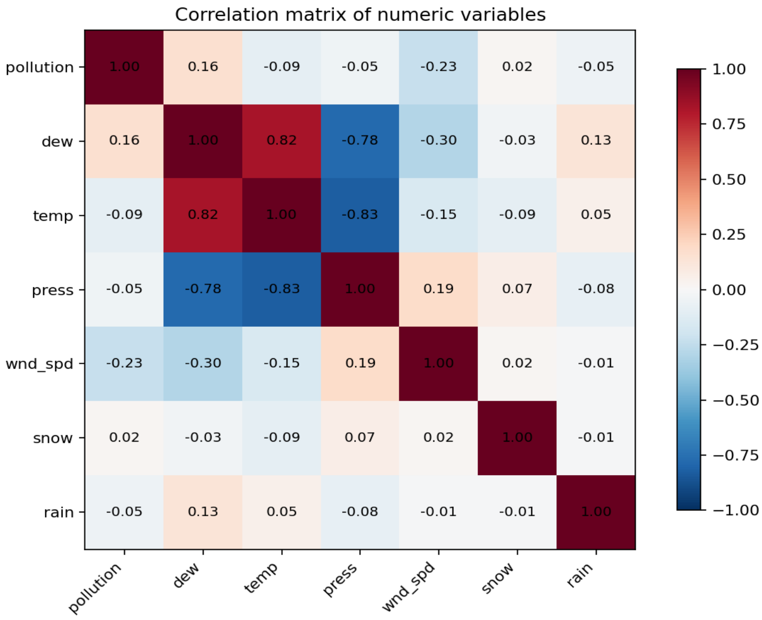

Figure 2: Correlation matrix of the numeric variables.

3. Models

Persistence baseline: predicts that the next PM2.5 value equals the previous hour. **Ridge regression:** tests whether a regularized linear model captures useful lag structure. **Random Forest:** models nonlinear thresholds and interactions through decision-tree ensembles. **Gradient Boosting:** builds an additive nonlinear regressor optimized by sequential residual correction. **MLP:** uses a feed-forward neural network on the same lagged feature vector. **LSTM:** uses a recurrent neural network designed for sequence data and predicts from the final hidden state.

4. Results

The supervised dataset contains 35016 training windows and 8760 test windows. The table below reports MAE, RMSE, and R2 on the chronological hold-out set. Lower MAE/RMSE and higher R2 indicate better forecasts.

Model	MAE	RMSE	R2
LSTM	12.339	23.060	0.939
MLP	12.697	23.346	0.938
Ridge	12.786	23.717	0.936
Random Forest	12.531	23.751	0.936
Persistence	12.430	24.515	0.931
Gradient Boosting	12.960	24.600	0.931

Table 1: Forecasting results on the temporal hold-out test set.

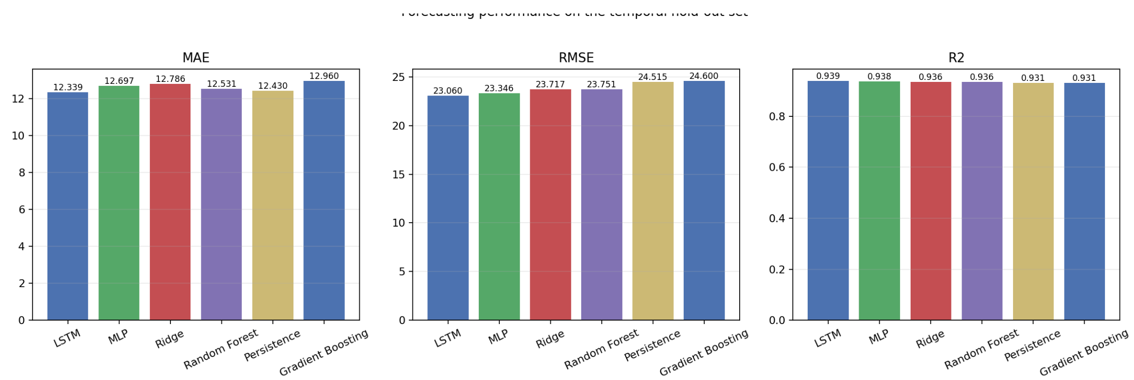

Figure 3: Metric comparison across all forecasting models.

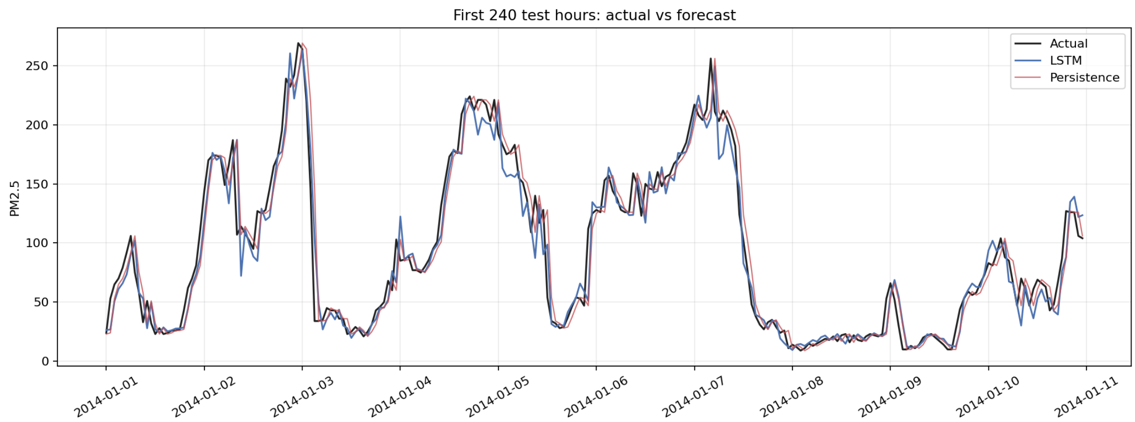

Figure 4: First 240 test hours: actual PM2.5, best model forecast, and persistence baseline.

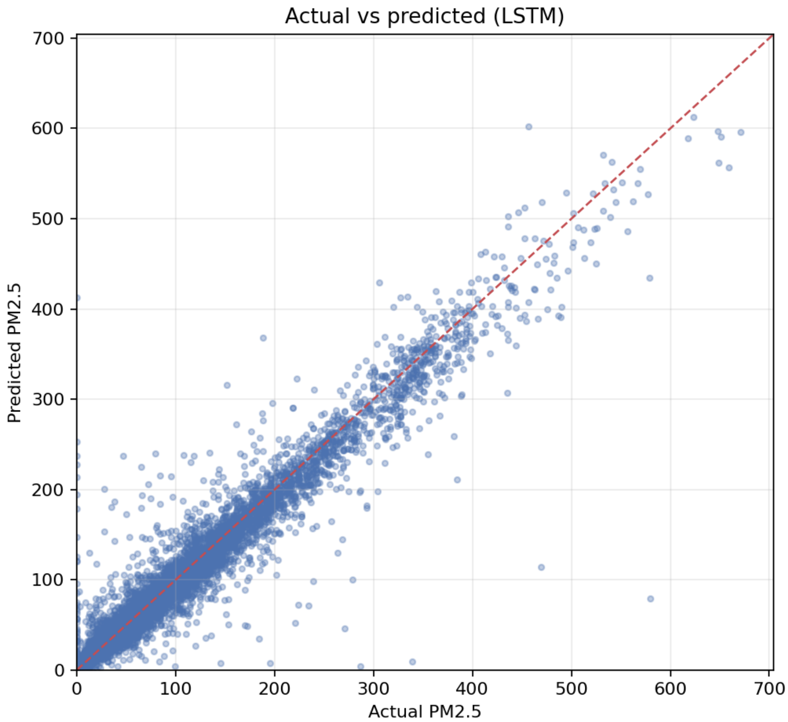

Figure 5: Actual vs predicted PM2.5 for LSTM.

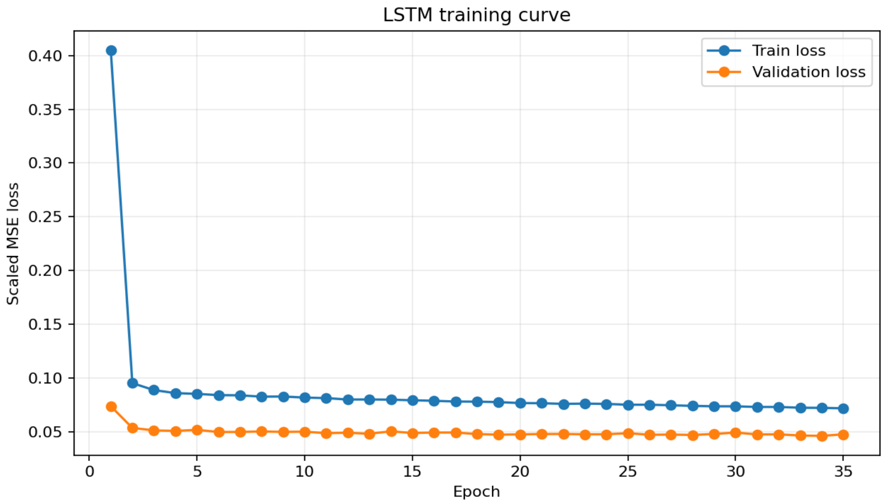

Figure 6: LSTM training and validation loss curves.

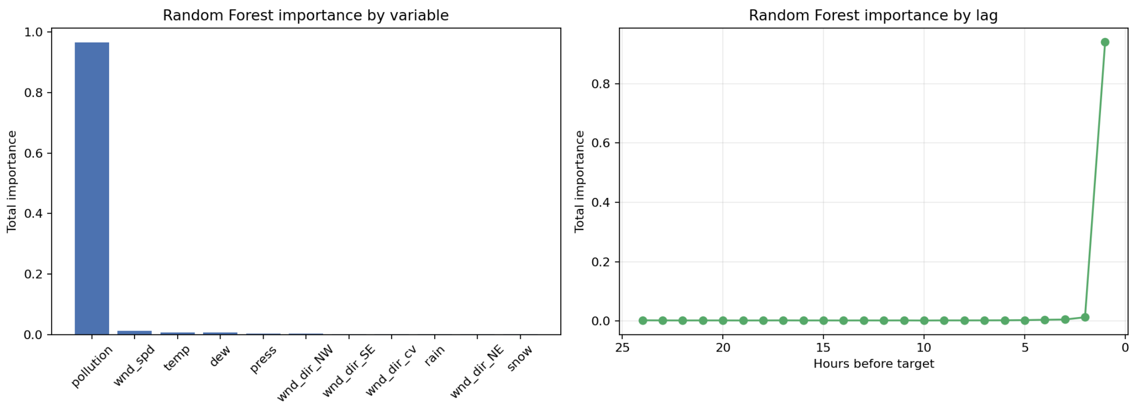

Figure 7: Random Forest lag-feature importance, aggregated by original variable and by hour-before-target.

5. Discussion

The persistence baseline is strong because air pollution changes smoothly over adjacent hours. Ridge regression improves or degrades depending on how linear the local dynamics are, while tree ensembles can model nonlinear interactions among recent pollution, wind, temperature, and pressure. MLP and LSTM add neural-network baselines: the MLP treats the lag window as one vector, whereas the LSTM explicitly processes the observations as a sequence.

The best result, LSTM, suggests that nonlinear temporal patterns are important for this forecasting task. The lag-importance plot also shows that the most recent pollution values dominate the prediction, which is consistent with the temporal autocorrelation expected in hourly air-quality data.

6. Conclusion

This experiment completes the multivariate air-quality forecasting pipeline: data loading, preprocessing, lag-window construction, model comparison, and quantitative evaluation. The chronological split keeps the evaluation realistic, and the included baseline helps distinguish genuine modeling gains from the natural persistence of PM2.5 values.

References

[1] C. M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
[2] F. Pedregosa et al., *Scikit-learn: Machine Learning in Python*, JMLR, 2011.
[3] Kaggle, LSTM Datasets: Multivariate and Univariate Air Quality Data.